

5. MODELLING IN STATE SPACE

Complex systems may have multiple inputs and multiple outputs and may be time varying. Because of the necessity of meeting increasingly stringent requirements on the performance of control systems, a new approach has been developed in the sixties. The advent of computers also helped its development. This is known as the modern control theory

The difference between the modern control theory and the conventional (classical) control theory is that the former is applicable to multiple-input-multiple-output systems which may be linear or nonlinear, time invariant or time varying, while the latter is applicable only to linear, time invariant, single-input-single-output systems. Also the modern control theory is essentially a time-domain approach, while the classical control theory is a complex frequency-domain approach.

The new approach is based on the concept of state.

State: The state of a dynamic system is the smallest set of variables called state variables such that the knowledge of these variables at $t = t_0$ together with the knowledge of the input for $t \geq t_0$ completely define the behaviour of the system for any time $t \geq t_0$.

State variables: The state variables of a dynamic system are the smallest set of variables that determines the state of the dynamic system. If n variables $x_1, x_2, \dots, x_n$ are needed to completely describe the behavior of a dynamic system, then such n variables are a set of state variables. Note that state variables need not generally be physically measurable or observable quantities.

State vector: If n state variables are needed to completely define the behavior of a given system, then these n state variables can be considered as the n component of vector $\mathbf{x}$. Such a vector is called state vector.

$$\mathbf{x} = [x_1 \ x_2 \ \dots \ x_n]$$

State space: The n -dimensional space whose coordinate axes consist of the x_1 axis, x_2 axis, $\dots, x_n$ axis, where $x_1, x_2, \dots, x_n$ are state variables, is called a state space.

In state space analysis we are concerned with three types of variables: input variables, output variables, and state variables. Also remember that the state space representation of a given system is not unique. But the number of state variables will be the same in different state space models of a given system.

Consider the second order system shown. We know the differential equation governing this system is $m\ddot{y} + b\dot{y} + ky = u$. It is a second order differential equation. This means that the system involves two integrators. Let us define the state variables $x_1(t)$, and $x_2(t)$

$$x_1(t) = y(t)$$

$$x_2(t) = \dot{y}(t)$$

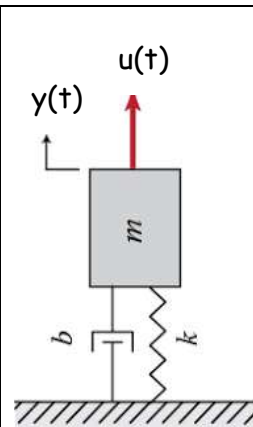
Then we obtain $x_2 = \dot{x}_1$

Then we can re-write the governing equation as $\ddot{y} = -\frac{b}{m}\dot{y} - \frac{k}{m}y + \frac{1}{m}u$

$$\dot{x}_2 = -\frac{k}{m}x_1 - \frac{b}{m}x_2 + \frac{1}{m}u$$

The output is

$$y = x_1$$



Now we have the state equations

$$\begin{aligned}\dot{x}_1 &= x_2 \\ \dot{x}_2 &= -\frac{k}{m}x_1 - \frac{b}{m}x_2 + \frac{1}{m}u\end{aligned}$$

The second order differential equation is converted into two first order equations.

We can write these equations in the vector-matrix form as

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -\frac{k}{m} & -\frac{b}{m} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 0 \\ \frac{1}{m} \end{bmatrix} u$$

The output equation is

$$y = [1 \quad 0] \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

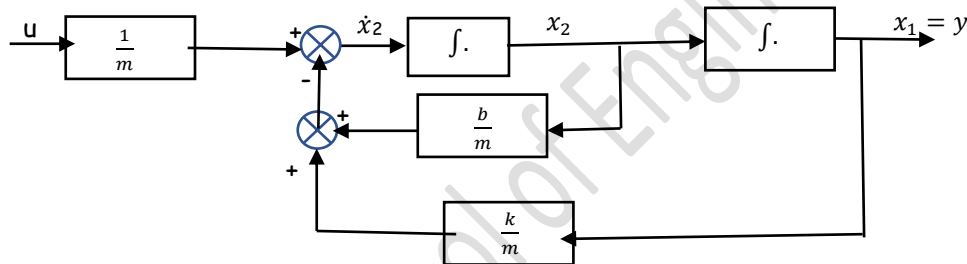
This can be written in the standard form as

$$\dot{\mathbf{x}} = [\mathbf{A}]\mathbf{x} + [\mathbf{B}]u$$

$$y = [\mathbf{C}]\mathbf{x} + [\mathbf{D}]u$$

where $[\mathbf{A}] = \begin{bmatrix} 0 & 1 \\ -\frac{k}{m} & -\frac{b}{m} \end{bmatrix}$, $[\mathbf{B}] = \begin{bmatrix} 0 \\ \frac{1}{m} \end{bmatrix}$, $[\mathbf{C}] = [1 \quad 0]$, $[\mathbf{D}] = 0$, $\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$ is the state vector, u is the input, and y is the output.

We can represent the state equations as block diagram shown below.



*Example

Obtain the state space representation of the system whose transfer function is

$$\frac{Y(s)}{R(s)} = \frac{24}{s^3 + 9s^2 + 26s + 24}$$

From the transfer function, we can write $Y(s)(s^3 + 9s^2 + 26s + 24) = 24R(s)$. The differential equation corresponding to this is $\ddot{y} + 9\dot{y} + 26y = 24r$

$$\ddot{y} = -9\dot{y} - 26y + 24r$$

The equation is of order three. Hence, there should be three state variables. Let us define the state variables as x_1, x_2, x_3 so that

$$y = x_1$$

$$\dot{x}_1 = x_2 = \dot{y}$$

$$\dot{x}_2 = x_3 = \dot{\dot{y}}$$

$$\dot{x}_3 = \ddot{y} = -9x_3 - 26x_2 - 24x_1 + 24r$$

That is,

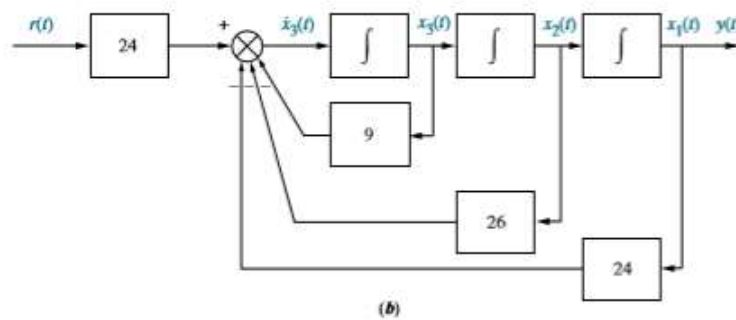
$$\dot{x}_1 = x_2$$

$$\dot{x}_2 = x_3$$

$$\dot{x}_3 = -24x_1 - 26x_2 - 9x_3 + 24r$$

And the output is $y = x_1$

The third order differential equation is reduced to three first order differential equations. These equations can be represented in a block diagram as shown below.



We can write this in matrix form as

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -24 & -26 & -9 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 24 \end{bmatrix} r$$

Output is $y = [1 \ 0 \ 0] \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$

This can be written in the standard form as

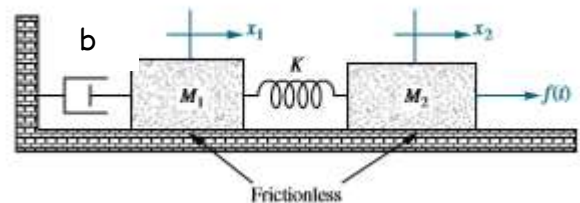
$$\dot{[x]} = [A][x] + [B]r$$

$$y = [C][x] + [D]r$$

where $[A] = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -24 & -26 & -9 \end{bmatrix}$, $[B] = \begin{bmatrix} 0 \\ 0 \\ 24 \end{bmatrix}$, $[C] = [1 \ 0 \ 0]$, $[D] = 0$, $[x] = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$ is the state vector, r is the input, and y is the output.

*Example

Obtain the state space representation for the system shown.



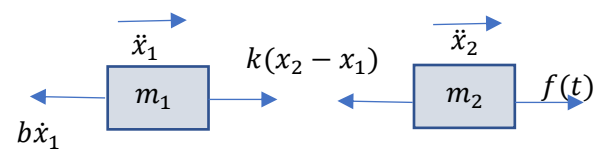
The equations of motion can be written as

$$m_1 \ddot{x}_1 = -b\dot{x}_1 + k(x_2 - x_1)$$

$$m_2 \ddot{x}_2 = f(t) - k(x_2 - x_1)$$

$$m_1 \ddot{x}_1 + b\dot{x}_1 + kx_1 - kx_2 = 0$$

$$m_2 \ddot{x}_2 + kx_2 - kx_1 = f(t)$$



Now we have two second order differential equations. Let us assume four state variables (y_1, y_2, y_3, y_4), two each for each equation. $y_1 = x_1$ and $y_3 = x_2$

$$\dot{y}_1 = y_2$$

$$\dot{y}_2 = -\frac{k}{m_1}y_1 - \frac{b}{m_1}y_2 + \frac{k}{m_1}y_3$$

$$\dot{y}_3 = y_4$$

$$\dot{y}_4 = \frac{k}{m_2}y_1 - \frac{k}{m_2}y_3 + \frac{1}{m_2}f(t)$$

In the matrix form

$$\begin{bmatrix} \dot{y}_1 \\ \dot{y}_2 \\ \dot{y}_3 \\ \dot{y}_4 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ -\frac{k}{m_1} & -\frac{b}{m_1} & \frac{k}{m_1} & 0 \\ 0 & 0 & 0 & 1 \\ \frac{k}{m_2} & 0 & -\frac{k}{m_2} & 0 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \\ y_4 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 0 \\ \frac{1}{m_2} \end{bmatrix} f$$

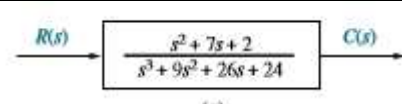
$$[\dot{y}] = [A][y] + [B]f$$

the outputs, $\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \\ y_4 \end{bmatrix}$

***Example**

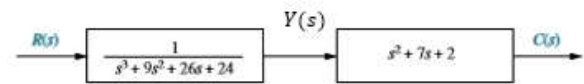
Obtain the state space representation of the system

whose transfer function is $G(s) = \frac{C(s)}{R(s)} = \frac{s^2+7s+2}{s^3+9s^2+26s+24}$



This problem is different from the earlier ones since the numerator has a polynomial in s instead of a constant.

Separate the system into two cascaded blocks as shown. The first block contains the denominator and the second one the numerator.



Let us find out the state space representation of the first block.

$$\frac{Y(s)}{R(s)} = \frac{1}{s^3 + 9s^2 + 26s + 24}$$

$(s^3 + 9s^2 + 26s + 24)Y(s) = R(s)$. By taking the inverse transform, the governing equation is

$$\ddot{y} + 9\dot{y} + 26y = r$$

$$\ddot{y} = -24y - 26\dot{y} - 9\ddot{y} + r$$

We can consider the state variables as x_1, x_2, x_3 where $x_1 = y$, $x_2 = \dot{x}_1$ and $x_3 = \dot{x}_2$. Then,

$$\dot{x}_3 = -24x_1 - 26x_2 - 9x_3 + r$$

The state equations are

$$\begin{aligned} \dot{x}_1 &= x_2 \\ \dot{x}_2 &= x_3 \\ \dot{x}_3 &= -24x_1 - 26x_2 - 9x_3 + r \end{aligned}$$

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -24 & -26 & -9 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} r$$

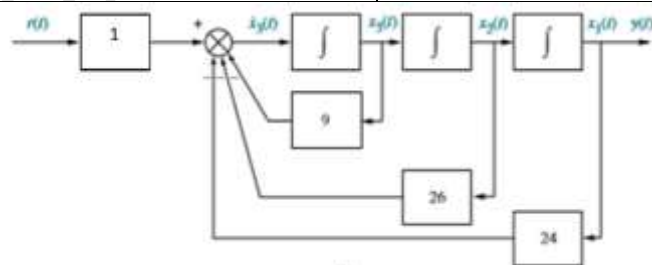
The output is $y = x_1$

$$y = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

This can be represented by the block diagram as shown. Now considering the second block, we can write $C(s) = (s^2 + 7s + 2)Y(s)$. That is,

$$\ddot{y} + 7\dot{y} + 2y = c$$

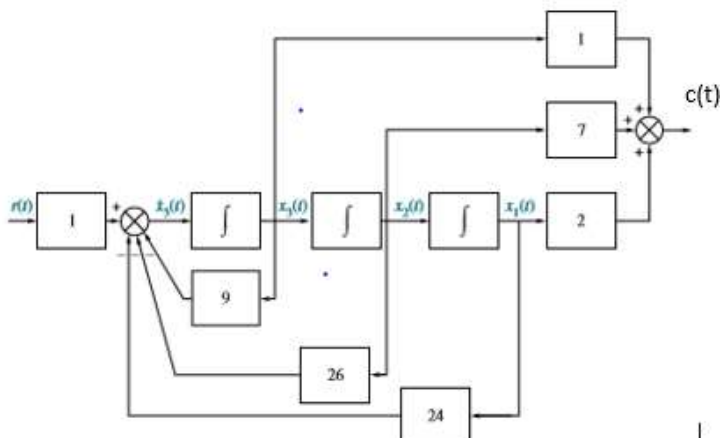
$$x_3 + 7x_2 + 2x_1 = c$$



The final output is

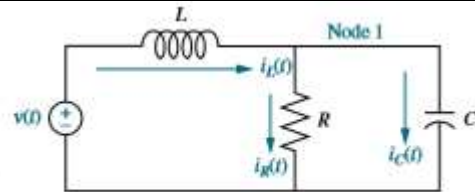
$$c = \begin{bmatrix} 1 & 7 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

The figure shows the block diagram representation of the complete state space model.



#Exercise 27

Obtain the state space representation of the electrical network shown. The output is $v_o(t)$



#Exercise 28

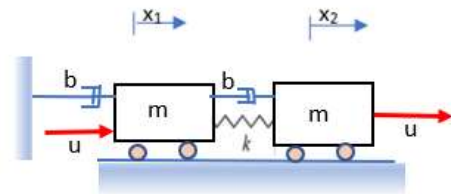
Obtain the state space representation of the system with transfer function

$$G(s) =$$

$$\frac{C(s)}{R(s)} = \frac{2s+1}{s^2+7s+9}$$

#Exercise 29

Obtain the state space representation of the system shown.



Correlation Between Transfer Functions and State Space Equations

Let us see how to derive the transfer function of a single-input-single-output system from the state space equations. Consider a system whose transfer function is given by $G(s) = \frac{Y(s)}{U(s)}$. The state space equations of this system can be written as

$$\dot{[x]} = [A][x] + [B]u$$

$$y = [C][x] + Du$$

Where $[x]$ is the state vector, u is the input, and y is the output.

Taking the Laplace Transforms of the above equations and setting all initial conditions to zero

$$s[X](s) = [A][X](s) + [B]U(s)$$

$$Y(s) = [C][X](s) + DU(s)$$

$$[X](s) \{sI - [A]\} = [B]U(s) \gg \gg \gg [X](s) = \{sI - [A]\}^{-1}[B]U(s)$$

$$Y(s) = [C]\{sI - [A]\}^{-1}[B]U(s) + DU(s) \gg \gg \gg Y(s) = \{[C]\{sI - [A]\}^{-1}[B] + D\}U(s)$$

$$G(s) = \frac{Y(s)}{U(s)} = [C]\{sI - [A]\}^{-1}[B] + D$$

Let us take the example of the spring mass system considered earlier. The state space representation of the system is

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -\frac{k}{m} & -\frac{b}{m} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 0 \\ \frac{1}{m} \end{bmatrix} u$$

The output equation is

$$y = [1 \quad 0] \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

where $[A] = \begin{bmatrix} 0 & 1 \\ -\frac{k}{m} & -\frac{b}{m} \end{bmatrix}$, $[B] = \begin{bmatrix} 0 \\ \frac{1}{m} \end{bmatrix}$, $[C] = [1 \quad 0]$, $[D] = 0$, $[x] = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$ is the state vector, u is the input, and y is the output. For simplicity, let us assume some numerical values for m , b , and k .

$$m=2, b=3, k=5. \text{ Hence, } [A] = \begin{bmatrix} 0 & 1 \\ -\frac{5}{2} & -\frac{3}{2} \end{bmatrix}, [B] = \begin{bmatrix} 0 \\ \frac{1}{2} \end{bmatrix},$$

$$G(s) = [C]\{sI - [A]\}^{-1}[B] + D = [1 \quad 0] \left\{ \begin{bmatrix} s & 0 \\ 0 & s \end{bmatrix} - \begin{bmatrix} 0 & 1 \\ -\frac{5}{2} & -\frac{3}{2} \end{bmatrix} \right\}^{-1} \begin{bmatrix} 0 \\ \frac{1}{2} \end{bmatrix}$$

$$= [1 \quad 0] \left\{ \begin{bmatrix} s & -1 \\ \frac{5}{2} & s + \frac{3}{2} \end{bmatrix} \right\}^{-1} \begin{bmatrix} 0 \\ \frac{1}{2} \end{bmatrix} = \frac{[1 \quad 0] \begin{bmatrix} s + \frac{3}{2} & 1 \\ -\frac{5}{2} & s \end{bmatrix} \begin{bmatrix} 0 \\ \frac{1}{2} \end{bmatrix}}{s^2 + \frac{3}{2}s + \frac{5}{2}} = \frac{[s + \frac{3}{2} \quad 1] \begin{bmatrix} 0 \\ \frac{1}{2} \end{bmatrix}}{s^2 + \frac{3}{2}s + \frac{5}{2}} = \frac{\frac{1}{2}}{s^2 + \frac{3}{2}s + \frac{5}{2}} = \frac{1}{2s^2 + 3s + 5}$$

$$G(s) = \frac{1}{2s^2 + 3s + 5} = \frac{1}{ms^2 + bs + k}$$

#Exercise 30

Convert the state and output equation to a transfer function

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} -4 & -1.5 \\ 4 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 2 \\ 0 \end{bmatrix} u$$

$$y = [1.5 \quad 0.625] \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

#Exercise 31

Obtain the state space representation of the system defined by

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} = \begin{bmatrix} -1 & 1 & 0 \\ 0 & -1 & 1 \\ 0 & 0 & -2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} r \quad \text{and} \quad y = [1 \quad 0 \quad 0] \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

#Exercise 32

Using Matlab/Scilab script, convert the given transfer function to state space representation

$$G(s) = \frac{2s^2 + 8s + 6}{s^3 + 8s^2 + 16s + 6}$$

#Exercise 33

Using Matlab/Scilab script obtain the unit impulse and unit step response of the following system

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -1 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u$$

$$y = [1 \quad 0] \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + [0]u$$

References

1. Modern Control Systems : Richard C. Dorf and Robert H. Bishop, 8/e, Addison-Wesley
2. Modern Control Engineering : Katsuhiko Ogatta, Pearson Education, LPE
3. Control Systems Engineering: Norman S. Nise, 6/e, John Wiley & Sons

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